



DO-IT-YOURSELF: PROJECT

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Once all functions are properly defined in the code, save it. Fig. 5 shows the source code snippet.

The water level is displayed in litres (0 to 15) and also as a percentage (0% to 100%). Graphically, the process is illustrated using a tank animation that updates progressively at 25%, 50%, 75%, and so on. The water fill colour changes to RED when the level is below 10%, and gradually shifts from CYAN to full BLUE when the water level exceeds 80%. Similarly, temperature is displayed in degrees Celsius (20°C to 70°C) with an animated

bar graph that reflects readings such as 35°C, 45°C, 60°C, and 70°C. The bar colour transitions from BLUE to CYAN to YELLOW to RED as the temperature increases.

Construction and testing

Before assembly, upload the programme by selecting the correct board and COM port. Then refer to the assembly diagram shown in Fig. 5 and Table 2 before assembling the system. Also, refer to the circuits shown in Figs. 3 and 4 during assembly. This system uses the Arduino Nano, but the Arduino Uno can also be used for the same purpose.

To calibrate and mount the ultrasonic distance sensor module HC-SR04 (S2), position it properly above the top level of the water so that the ultrasonic waves can travel unobstructed for accurate measurement.

To test the system, after proper assembly, power the device either

through the USB port or using a 5V battery. Once powered, the data will be displayed on the LCD screen. As shown in Fig. 6, the water level in the tank is displayed in both litres and percentage, representing the actual water level. This is accompanied by a real-time, colour-changing animation of the tank filling with water. Similarly, the actual water temperature is shown in degrees Celsius with a dynamic, colour-changing bar graph.

This system provides a visually rich and effective method for monitoring water level and temperature using multicoloured graphics and animation. Fig. 6 shows the display of water level and temperature on the TFT. **EFY**

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